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Inattentional Blindness in Augmented Reality Head-Up Display-Assisted Driving

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Department of Psychology, Zhejiang Sci-Tech University, Hangzhou, China

ABSTRACT

Augmented reality head-up display (AR HUD) is a new technology in assisted driving, which can add extra information to the driving environment in real-time to help the driver better perceive road situation. AR HUD can enhance driving safety but may also encourage inattentional blindness. Hence, this study aims to examine whether AR HUD induces inattentional blindness and determine whether workload intensifies their relationship. In experiment 1, 60 participants were randomly assigned to three groups and watched three types of augmented reality (AR)-augmented driving videos, respectively. They were instructed to respond to any critical events, but only their responses to road-crossing pedestrians were recorded. Results show that AR HUD reduces inattentional blindness when pedestrians are augmented but encourages inattentional blindness when pedestrians are not augmented. In experiment 2, 20 participants viewed AR-augmented driving videos of high and low workloads. Pedestrians were not augmented in all videos. Result reveals that a high workload induces more inattentional blindness than low workload. The finding confirms that AR HUD induces inattentional blindness, and a high workload will intensify this relationship. The future design of the AR HUD assisted-driving system should consider the risk of inattentional blindness and come up with corresponding countermeasures.

1. Introduction

Augmented reality (AR) technology refers to a technology in which virtual information is added to the real environment through engineering and computer technologies. In addition, the real and virtual environments are synchronized in the same spatial relationship in real-time (Moussa et al., 2012). With the increasing maturity of AR technology, AR has been applied increasingly to everyday life, such as games, assembly, navigation, teaching, and medical surgery, offering substantial convenience for work, life, and entertainment. Studies in driving tried to combine AR technology and head-up display to create a novel approach to present information to drivers. Two types of AR HUD design exist according to the driver's view perspectives, that is, screen-fixed AR HUD and world-fixed AR HUD, which are used for to different purposes. Screen-fixed AR HUD is suitable for presenting driving-related information not directly attached to real-world objects such as vehicle speed, and world-fixed AR HUD is suitable for augmenting real-world road objects and can be applied in wayfinding and hazard warning systems (Gabbard et al., 2014). However, when applying world-fixed AR HUD to enhance road objects, the augmented objects may attract too much attention from drivers causing them to ignore "unpredictable events" (e.g., the sudden appearance of pedestrians). In this case, the system failed to augment unpredictable stimulus in time, thereby, inducing inattentional blindness.

Inattentional blindness is a phenomenon that the observer's attention is shifted to other places, so even when looking at them, the observer fails to detect certain evident stimuli,

which happens frequently and sometimes threatens people's safety (Neisser, 1979; Neisser & Becklen, 1975). The AR technology application is often accompanied by inattentional blindness. Dixon et al. (2014, 2013) found that doctors using AR navigation technology experience more inattentional blindness during surgery than those who do not. AR technology is also widely applied in a driving-assistance device. However, AR technology is not yet fully mature in the assisted driving field, and its advantages and disadvantages need further research and discussion. Special focus should be put on AR-induced inattentional blindness, which has been a major concern in the application of AR technologies and is closely related to road safety. Many studies in cognitive psychology found that the individual's attention to the main task is the main cause of inattentional blindness. Moreover, researchers already discovered several factors that affect inattentional blindness (Mack & Rock, 1998; Most et al., 2001; Simons & Chabris, 1999), including the location, shape, color, and size of unpredictable stimuli (Hyman et al., 2010; Most et al., 2001; Simons & Chabris, 1999), the eye movement effects (Beanland & Pammer, 2010b), previous experience of drivers (Beanland & Pammer, 2010a), and the difficulty of the main task (Cartwright-Finch & Lavie, 2007; Lavie, 1995). Some studies also found that although the incidence rate is relatively low, individuals still experience inattentional blindness even without the main task. This finding further indicates that the features of unpredictable stimuli can exert an influence on inattentional blindness (Davies et al., 2013; White & Caird, 2010). With regard to driving, although

driving shares the same theoretical basis with the cognitive study on inattention blindness, considerable differences still exist. Driving requires the drivers to control the car, constantly pay attention to the surrounding environment, and adjust the trajectory and speed of the vehicle according to the surrounding environment. Each task is associated with driving safety and cannot be defined as the main task. That is to say, a driving task may be much more difficult than the full attention task in the cognitive study. In the inattention blindness paradigm used in cognitive studies, participants only need to detect certain stimuli while performing rather simple main tasks (Mack & Rock, 1998; Most et al., 2005, 2001). For example, in Mack and Rock (1998) study, participants needed to detect a simple geometric shape (unexpected stimuli) while discriminating the longer arm of a cross (main task). Moreover, Most et al. (2001) showed crosses on the screen (unexpected stimuli) while the participants were instructed to count the number of the letters showed on the screen simultaneously (main task).

1.1. Inattention blindness in driving

Inattention blindness during driving may cause serious consequences, and thus, this topic has gained considerable attention from researchers. Existing studies related to blindness in driving mainly focused on the condition and factors that lead to inattention blindness and were committed to eliminating such phenomenon in driving. With the evolution of automobile equipment and technology, new study topics and methods continue to emerge.

Romer et al. (2014) proposed a model of sources of attention failures in novice drivers, which differentiates distraction and inattention blindness based on whether driving-related tasks cause ignorance of road conditions. This model reveals that distraction is an attention failure caused by the driver's attention shifting to other non-driving-related tasks, such as using mobile phones and talking. However, inattention blindness means that the driver's attention is focused on the driving task but still fails to notice key stimuli. White and Caird (2010) first distinguished looked-but-failed-to-see error (LBFTS) error from inattention blindness. LBFTS error refers to a common phenomenon in driving that the driver looked at the target stimulus but failed to detect it, that is, he did not actually "see" it. They found that conversations during driving, particularly with attractive passengers, are more likely to trigger LBFTS, and extroverted drivers are more likely to experience LBFTS. Given that the driver's sight is already on the target object when LBFTS occurs, they speculate that conversational distraction should be cognitive and has nothing to do with the gaze situation. They thus believe that LBFTS should be regarded as a special kind of inattention blindness that needs further exploration.

Early studies on inattention blindness in driving normally adopted the conventional dual-task paradigm. They used a task that is non-related to driving tasks or even interfering with driving as the main task, such as using a mobile phone or talking to passengers (Strayer et al., 2003; White & Caird, 2010). In Strayer's experiment, no significant difference is observed in eye movement data between participants who

used a mobile phone while driving and those who did not. However, participants' memory toward specific objects who used a mobile phone was significantly reduced, indicating that using mobile phones can cause more inattention blindness (Strayer et al., 2003).

Many researchers also studied the inattention blindness of drivers while only completing driving tasks. Most and Astur (2007) conducted the first well-recognized study that explored inattention blindness in driving. They found that if the color of the unpredictable stimuli is different from that of the stimuli the participant pays attention to, then the participant is more likely to experience inattention blindness, and the response time is longer even when the participant notices the stimulus (Most & Astur, 2007). This finding is consistent with the conclusion drawn in a cognitive psychology study (Simons & Chabris, 1999).

In the studies of inattention blindness in driving, the self-report of participants is one of the most common indicators, such as the score of recall, recognition, or forced-choice test (H. Kim & Gabbard, 2019; Pammer & Blink, 2013). However, self-report data may be lacking in investigating the influence of inattention blindness on driving performance. Therefore, many studies adopted behavioral indicators. In the study of inattention blindness in the driving, reaction time is introduced through driving experiments using static mode (i.e., picture mode), video mode, and simulators as objective indicators, including pressing a certain key when observing unpredictable stimulus or performing specific driving operations such as turn on lights or windshield wipers (Koivisto & Revonsuo, 2008; Murphy & Greene, 2015; Pammer et al., 2015; Pammer & Blink, 2013; Yanko & Spalek, 2013).

1.2. Influencing factors of inattention blindness in driving

The types and properties of unpredictable stimuli will affect inattention blindness in driving. A study demonstrated that in driving, different unpredictable stimuli may result in a different incidence of inattention blindness (Pammer et al., 2015). This finding is similar to the study of Mack and Rock (1998) whose result shows that the participants have a different sensitivity to different stimuli. Pammer and Blink (2013) also concluded that the consistency of accidental stimuli and the environment will affect inattention blindness. The unpredictable stimuli frequently used in the recent study include pedestrians (Murphy & Greene, 2015, 2016; Pammer et al., 2015; Pammer & Blink, 2013; Yanko & Spalek, 2013), motorcycles (Most & Astur, 2007), vehicles (Murphy & Greene, 2016; Yanko & Spalek, 2013), road signs (Kennedy & Bliss, 2013), and non-safety-related stimuli (Remington et al., 2014; Strayer et al., 2003). Among them, pedestrians are the most used stimuli. Pammer and Blink (2013) studied the impact of consistency between accidental stimuli and driving environment on inattention blindness. Results show that in the urban environment, participants are more likely to experience inattention blindness for kangaroos that do not match the environment than businessmen. On the contrary, in a rural environment, no significant difference is found between the two stimuli (Pammer & Blink, 2013).

The experiment revealed that the consistency of accidental stimuli and the driving environment may affect the occurrence of inattention blindness.

High workloads can also lead to an increase in inattention blindness. Kennedy and Bliss (2013) showed that even when only completing high-load driving-related tasks, participants are very prone to inattention blindness. Murphy and Greene (2015, 2016) also studied inattention blindness in driving under different workloads via driving simulators and manipulating workloads through a target searching task and narrow gap driving task respectively. They found that drivers are more likely to experience inattention blindness under high perceptual load because under the low perceptual load, the drivers paid additional attention to driving-related stimuli. By contrast, under the high perceptual load, the driver paid equal attention to driving-related and unrelated tasks. Their study also proved that no significant difference exists in the occurrence of inattention blindness between unpredictable stimuli that appeared the first and second time. This result indicates that cognitive load, and not participants' expectation, affects inattention blindness.

However, the way to manipulate workload may lead to inconsistent results. For example, road familiarity is one of the most unique factors of inattention blindness in driving. Researchers proposed that driving on familiar roads will encourage automated processing of road information that leads to low workload, some studies showed that driving on familiar roads leads to more inattention blindness (Charlton & Starkey, 2013; Yanko & Spalek, 2013). This conclusion accords with De Fockert and Bremner (2011), in which low workload may also cause inattention blindness. However, these results are contrary to that drawn by Kennedy and Bliss (2013) and Murphy and Greene (2015, 2016). In the present study, we speculate that this discrepancy of results may be attributed to the various manipulations of the driver's workload (e.g., road familiarity in Charlton and Starkey (2013) and Yanko and Spalek (2013), subjective workload after completed the driving task in Kennedy and Bliss (2013), and driving through car gaps of different widths in Murphy and Greene (2015, 2016)). Hence, we used road complicity to induce different workloads and asked participants to complete workload scales after completing all tasks as supporting information in our study to ensure that we successfully manipulate participants' workload.

1.3. AR HUD and inattention blindness

The AR assistance in driving can be divided into visual, auditory, and tactile support according to the different sensory channels, and all forms have been put into use in certain areas of driving (S. Kim & Dey, 2016). Most of the driving information is obtained through visual perception. Visual perception can accurately seize long-distance information and the direction, trajectory, and speed of the vehicle. Nowadays, the AR HUD is widely used in AR-assisted driving. In addition, AR HUD integrates AR information in the conventional HUD by projecting real-time driving information on the HUD, thereby providing the driver with additional information from the first perspective (Ng-Thow-Hing et al.,

2013). Moreover, AR HUD can enhance the driver's visibility of important events on the road and may also enhance the driver's situational awareness. AR HUD is also suitable for drivers of different ages (Ng-Thow-Hing et al., 2013). Hwang et al. (2016) proved that AR HUD can enhance the driver's risk perception and reduce the driver's anxiety level.

However, AR HUD has certain disadvantages. When HUD was first applied in the aerospace field, many studies have found that HUD can bring about a phenomenon named "attention tunnel" (Karar & Ghosh, 2012). Attention tunnel refers to missed events caused by a slight mismatch of attention between the HUD and the external scenery (Karar et al., 2015). This phenomenon can be regarded as inattention blindness in driving. One view is that HUD may separate the attention of pilots from the real environment. In other words, pilots tend to ignore unpredictable stimuli when paying attention to the information on HUD (Moodi, 1995).

A similar phenomenon exists in the field of automobile driving. Gabbard et al. (2014) proposed that when driver pays additional attention to the information displayed on the windshield by HUD, they tend to ignore information in the real environment. They suggested that the role of AR HUD in driving has three levels: (1) guiding the driver to perceive the surrounding environment and operate the vehicle, including wayfinding, heading control, speed control, and distance control with other vehicles and objects; (2) guiding the driver to operate some functions that are more relevant to the driving task, such as turning on the turn signal lights and adjusting the wiper; (3) guiding the driver to complete the adjustment and entertainment functions, such as adjusting the air conditioning, information display, and other entertainment functions (Gabbard et al., 2014). The importance of these three levels decreases in order. Gabbard et al. (2014) suggested that if the second and third levels of AR HUD attract excessive attention from the drivers, then they can easily ignore the stimuli in the real driving environment, resulting in safety hazards.

Although previous studies provided evidence that AR HUD introduces inattention blindness, most of them focused on "distraction" (Romer et al., 2014). That is, non-driving-related tasks in AR HUD-assisted driving cause drivers to ignore unpredictable stimuli.

The factors that cause attention failure in the HUD may also cause attention failure in the AR HUD. However, few related studies focused on the factors influencing inattention blindness in AR HUD. Oh et al. (2016) suggested that the superposition of HUD on unpredictable stimuli in the real world may cause additional attention tunnel, that is, more inattention blindness. Some researchers were committed to improving AR HUD to make it a better assistant for drivers. H. Kim et al. (2016) provided the drivers with the distance information between the vehicle and the critical event through AR HUD, which effectively reduced inattention blindness. Then, Schwarz and Fastenmeier (2017) proved that using AR technology to provide the subjects with visual information by describing the accident ahead in detail can reduce the subject's time to shift their attention back to the road. In a recent study, H. Kim and Gabbard (2019) used driver's gaze behavior and situational awareness as quantifiable indicators of inattention blindness in AR HUD-assisted driving, providing a new method to evaluate the AR HUD interface.

They concluded that using a conformal graphic (e.g., a bounding box) to augment stimulus will encourage inattention blindness toward non-augmented environmental elements. However, Kim and Gabbard only augmented pedestrians in their study, excluding other important safety-critical elements, such as road signs. Unpredictable stimuli, such as pedestrians, are easily omitted by the system due to their unpredictability and current technical restraints. Thus, future studies need to explore conditions when unpredictable stimuli are not augmented in AR HUD-assisted driving to come up with corresponding countermeasures.

In summary, AR HUD should try to provide drivers with more appropriate and efficient information to prevent traffic accidents. However, many studies showed that AR HUD may lead to more inattention blindness. At present, only a few studies focused on the blindness caused by HUD, and even less research explored the impact of AR HUD on inattention blindness. The AR bounding box used in AR HUD can direct the driver's attention. This bounding box is designed to attract the driver's attention to the augmented stimuli, which may induce inattention blindness toward non-augmented stimuli (i.e., unpredictable stimuli in some situations). Moreover, existing studies showed that the characteristic of stimuli and cognitive load are associated with inattention blindness. However, no study has examined the effect of cognitive load on inattention blindness in the context of AR HUD display. Therefore, the influence of augmented condition and cognitive load on the occurrence of inattention blindness in the AR HUD driving task needs to be explored.

The present study focuses on inattention blindness in AR HUD-assisted driving tasks by exploring its forming condition and influencing factors through two experiments. The present study mainly aims to improve the AR HUD design and reduce inattention blindness. The objectives of the study are as follows:

- (1) Explore whether the application of AR-assisted display in-car driving will induce more inattention blindness. In experiment 1, some contingencies (i.e. the unpredictable event that has not been augmented by AR) were added in driving routes to induce the potential inattention blindness when the drivers use the AR HUD during driving.
- (2) Explore whether cognitive load affects the occurrence of inattention blindness in AR HUD-assisted driving. In experiment 2, we set two driving scenarios with different complexities of the driving environment to induce different levels of cognitive load in driving.

Accordingly, this study proposes the following hypothesis: Drivers using AR HUD without augmented pedestrian (AR HUD-NP) will experience the highest level of inattention blindness in a higher workload condition.

2. Experiment 1

2.1. Objectives

This study aims to explore whether AR HUD is used and whether unpredictable stimuli are augmented by AR HUD will increase inattention blindness in driving tasks. In this

experiment, the driving situations were divided into two types: with AR HUD and without AR HUD. Considering the large uncertainty of unpredictable stimuli, the existing AR HUD systems may not detect and augment the unpredictable stimuli in time. The unpredictable stimuli refer to the events that occur unpredictably to the drivers, such as the sudden appearance of a pedestrian. Thus, AR HUD conditions were further divided into AR HUD with and without augmented unpredictable stimuli conditions to investigate the potential effect of different augmented conditions of AR HUD on driving inattention blindness.

Thus, this experiment explored the effects of augmented conditions, including No AR HUD, AR HUD with augmented unpredictable stimuli, and AR HUD without augmented unpredictable stimuli, on inattention blindness in driving.

2.2. Methods

2.2.1. Participants

We recruited 60 licensed drivers with normal to corrected-to-normal vision. The average age of the participants was 20.94 years old ($SD = 2.32$), their average driving experience is 3.15 years ($SD = 1.35$), and their average mileage is 362 km. All participants provided written informed consent and were compensated for their participation.

2.2.2. Experimental equipment

This study used a driving simulator consisted of a driving simulator software (STISIMDRIVE M100K, Systems Technology Inc., Hawthorne, CA, USA), an operating system (Logitech MOMO, Newark, CA, USA) with a steering wheel, an accelerator, and brake pedals, two computers, and an INTECH IWB 72-inch LED screen with $1,920 \times 1,080$ px resolution is used in our study. The LED screen was approximately 1.5 m away from participants vertically.

All virtual sceneries were created on the STISIM 3 program and were projected on the LED screen. An SMI RED-m portable eye tracker was used to record participants' eye movements at a sampling frequency of 30 Hz. The eye movement data were analyzed using the SMI system software iView 3.5 and BeGaze 3.5.

The laboratory is an independent room of 25 m². The simulation car is set in the center of the room. The windshield is installed to make the driving environment more realistic. The cardboards are installed on both sides of the seats, and the LED screen is placed near the wall.

2.2.3. Materials

In the present study, we used driving videos generated with a medium-fidelity driving simulator to create a naturalistic driving experience. However, we cannot directly augment target stimuli in the driving simulator due to technical limitations. Thus, we adopted video materials used by Oh et al. (2016). To create the video footage, a licensed experimenter completed the driving tasks in the SITSIM driving simulator, and we captured the video footage through screen recording.

We created 10 videos in total. In the videos, the ego-car drove on a 2 km-long four-lane motorway at a constant speed of 65 km/h. We added various road elements, such as curves,

interactions with traffic lights, buildings, trees, road signs, vehicles, and pedestrians, to the scenarios to create a natural and realistic driving experience for drivers. We also added four types of critical events to the scenarios: (1) speed of the ego-car exceeding the speed limit (occurred two times), (2) the ego-car running a red light (occurred 2 times), (3) the ego-car driving too close to lead vehicles (occurred two times), and (4) pedestrian in front of the ego-car crossing the road (occurred six times). In all videos, the ego-car will not avoid any critical events. However, no collision would occur according to the system setting. All road elements and road events appeared at a random distance on the road.

2.2.4. Experimental variables and design

This experiment was a single-factor between-subject design in which three groups of participants need to experience different augmented conditions. The independent variable was augmented condition: (1) No AR HUD condition, (2) AR HUD without augmented pedestrian (AR HUD-NP) condition, (3) AR HUD with augmented pedestrian (AR HUD-P) condition. In this study, we use the No AR HUD condition as a baseline for driver behavioral and eye movement measurements.

To create a prototype of AR interfaces, we used a video-editing tool to overlay bounding boxes (i.e., the AR graphics) atop the driving videos. Table 1 shows the designing detail and demonstration graph of each augmented condition.

The dependent variables were the occurrence of inattention blindness, types of inattention blindness, reaction time without blindness, and subjective workload. We adopted White and Caird (2010) operational definition of inattention blindness, that is, drivers do not respond when the pedestrian is in the center of the windshield. Then we divided inattention blindness into ordinary blindness and looked-but-failed

-to-see error (LBFTS error) according to eye movement data (White & Caird, 2010) (Table 2). Ordinary blindness means that the participant neither looked at the pedestrian (gaze dwell time under 100 ms) nor responded to it. LBFTS error means that participants looked at pedestrians (gaze dwell time is and above 100 ms) without responding to it. The sampling frequency of the eye tracking device is 30 Hz, so 100 ms is approximately three consecutive video frames after pedestrians were in the center point of the windshield. The reaction time without blindness was defined as the interval time between the onset of the pedestrian's movement and the participant pressing the button. The workload is measured using the NASA Task Load Index (NASA-TLX) scale (Hart & Staveland, 1988), which includes six dimensions of workload (overall performance, frustration, effort, mental, physical, and temporal demand of a task). We used a weighted average TLX score as the workload indicator.

Pammer et al. (2015) noted that the types of unpredictable stimuli will affect inattention blindness. Therefore, we only used pedestrians as unpredictable stimuli in our study. The Governors Highway Safety Administration (GHSA) reported that the number of pedestrian fatalities in the United States has grown sharply in recent years (Governors Highway Safety Association, 2019). Thus, GHSA calls for engineering countermeasures to reduce pedestrian fatality. A report by Britain's Department of Transport shows that drivers failing to detect pedestrians accounted for 15% of pedestrian fatality (Department for Transport, 2019). Moreover, the current AR HUD designs have not reached a consensus on whether and how to augment pedestrians (Hwang et al., 2016; Karar et al., 2015; H. Kim & Gabbard, 2019; Langlois & Soualmi, 2016; Park & Kim, 2013; Park et al., 2013). Therefore, we choose pedestrians as our unpredictable stimuli.

Table 1. Designing details and demonstration graphs of three augmented conditions.

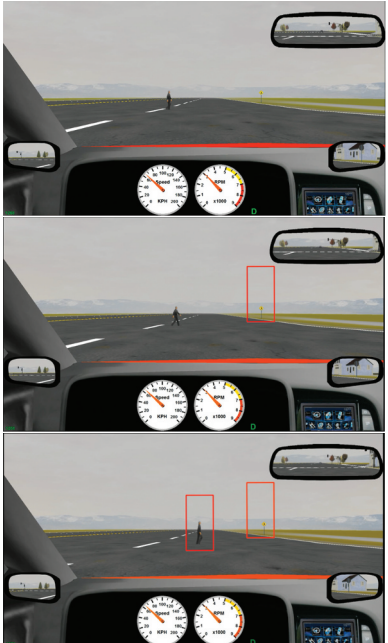
Types of augmented condition	Designing detail	Demonstration graph
No AR HUD	We did not augment anything in this condition.	
AR HUD without augmented pedestrian (AR HUD-NP)	We augmented road signs, traffic lights, and lead vehicles with bounding boxes, but we did not augment unpredictable pedestrians in this condition.	
AR HUD with augmented pedestrian condition (AR HUD-P)	We augmented road signs, traffic lights, lead vehicles, and unpredictable pedestrians in this condition.	

Table 2. Classification of no blindness, ordinary blindness, and LBFTS error.

		Response	
		Yes	No
Gaze dwell time	≥ 100 ms	Look and respond	LBFTS error
	< 100 ms	Not look but respond	Ordinary blindness

No blindness = Look and respond + Not look but respond; Inattentional blindness = LBFTS error + Ordinary blindness.

2.2.5. Procedure

The participants read and signed an informed consent form and completed a demographic questionnaire. The experimenter helped participants wear and calibrate the eye tracker using a three-point calibration mode of eye-tracking device. That is, participants gazed at three random locations in the front view, and the experimenter clicked the corresponding locations in the eye-tracking monitor to adjust the gaze point.

The experimenter also helped participants adjust the height of the chair to ensure that their eye sights were right in the center of the HUD display (the experimenter can observe participant's eyesight through the eye-tracking monitor).

We told the participant to imagine being the driver when watching these first-person-view videos. We also instructed them to press the designated buttons whenever they noticed a critical event in the videos, such as overspeeding, running a red light, being too close to the lead vehicle, or pedestrians crossing the road. We required participants to respond to all critical events because we want to prevent them from anticipating the pedestrians. That is, participants needed to press the button when they noticed the following: (1) the ego-car exceeded the speed limit, (2) the ego-car ran a red light, (3) the distance between the ego-car and the lead car was too close, (4) a participant in front was crossing the road. However, only the data of participants responding to pedestrians were used for analysis.

The participants first completed two practice trials. In the practice trials, all road settings were the same as the formal experiment. Moreover, critical events, such as overspeeding, running a red light, and leading vehicle decelerating (i.e., being too close to leading vehicles), were all included. However, unpredictable stimuli (i.e., pedestrians crossing the road) are excluded.

After practice, participants were randomly assigned to three groups and they were to watch three types of AR-augmented driving videos, respectively. Each group completed 10 formal trials (i.e., viewed 10 videos). Among these, six trials contained pedestrians' road-crossing events. The presentation order of 10 videos was counter-balanced by Latin squares. When participants complete one trial, the next trial automatically started on the basis of the system setting.

After completing all experimental trials, the participants were asked to complete the NASA-TLX scale to measure their overall workload of completing the entire experiment.

2.3. Results

2.3.1. Data processing

The occurrence of inattentional blindness is based on drivers' response (response or no response). We coded the result as no

blindness if participants responded to unpredictable pedestrians and inattentional blindness if they responded. The type of inattentional blindness was calculated based on eye movement data. We used unpredictable stimuli (i.e. pedestrians) as the area of interest (AOI) to analyze the gaze dwell time of the participants. We coded the result as LBFTS blindness if their gaze dwell time on the AOI was longer than three frames (100 ms at 30 Hz sample frequency) and they failed to press the button to respond (White & Caird, 2010). We coded the result as the ordinary inattentional blindness if the participants' gaze dwell time on the AOI was less than three frames and they failed to press the button (Table 2). Reaction time without blindness equals the exact time the participants press the brake pedal minus the exact time the unpredictable showed up. When analyzing the brake time pedal, we excluded the data when the participant did not press the brake pedal. For the NASA-TLX score, we asked the participants to finish two parts of the questionnaire. We multiplied the weighted score (the score derived from the second part of the questionnaire) and the scale score (the score derived from the first part of the questionnaire) and divided it by 15 to obtain our final workload score.

2.3.2. Occurrence of inattentional blindness

This experiment had 348 valid trials where we observed inattentional blindness in 168 trials. However, we excluded three trials, accounting for 1.78% of the total inattentional blindness trial, as their eye movement data were missing due to technical problems. Thus, we had 165 valid inattentional blindness trials for further analysis.

IBM SPSS Statistics 26.0 was used to analyze the experimental data. In the present study, the occurrence of inattentional blindness and the type of inattentional blindness were discrete and non-linear data. Thus, we adopted the generalized linear model (GLM) for data analysis. GLM is suitable for discrete data as it allows using a link function to connect the linear model to the non-normally distributed response variable. To reach consistency in data analysis, we also used GLM to analyze reaction time and NASA-TLX data.

We used the occurrence of blindness as the dependent variable and the augmented condition as the factor to perform GLM. We used binary logistic regression as the link function. The result showed that the effect of augmented condition on the occurrence of inattentional blindness was significant ($\chi^2(2) = 21.484, p < .001$). The GLM parameter estimates results showed that compared with the AR HUD-P condition, the No AR HUD ($\chi^2(1) = 6.554, p < .01$) and the AR HUD-NP conditions ($\chi^2(1) = 21.453, p < .001$) significantly increased the incidence of inattentional blindness (Table 3). We further conducted the GLM post hoc test to compare the incidence of inattentional blindness between AR HUD-NP and No AR HUD conditions (we used Bonferroni correction for multiple comparisons). The result showed that the AR HUD-NP condition had a marginally higher proportion of inattentional blindness No AR HUD condition ($p = .087$) (Figure 1a).

2.3.3. Type of inattentional blindness

We used the type of blindness as the dependent variable and the augmented condition as the factor to perform GLM. We

Table 3. Effect of augmented condition on behavior performance and NASA-TLX score.

Dependent Variable	Augmented condition	<i>B</i>	<i>SE</i>	<i>Wald</i> χ^2	<i>p</i>	<i>OR</i>	95% <i>CI OR</i>
The occurrence of inattentive blindness	No AR HUD	−0.697	0.272	6.554	.010**	0.498	[0.292, 0.849]
	AR HUD-NP	−1.275	0.275	21.453	.000***	0.280	[0.163, 0.479]
	AR HUD-P						
Type of inattentive blindness	No AR HUD	0.936	0.451	4.303	.038*	2.549	[1.053, 6.170]
	AR HUD-NP	1.394	0.431	10.487	.001***	4.032	[1.734, 9.377]
	AR HUD-P						
Reaction time without blindness	No AR HUD	111.655	31.4158	12.632	.000***	3.10E+48	[5.62E+21, 1.71E+75]
	AR HUD-NP	18.5	35.7514	0.268	.605	1.08E+08	[4.01E−23, 2.93E+38]
	AR HUD-P						
NASA-TLX	No AR HUD	6.061	4.1977	2.085	.149	428.733	[0.115, 1.60E+06]
	AR HUD-NP	−3.53	4.1977	0.707	.400	0.029	[7.83E−06, 109.633]
	AR HUD-P						

* $p < .05$; ** $p < .01$; *** $p < .001$.

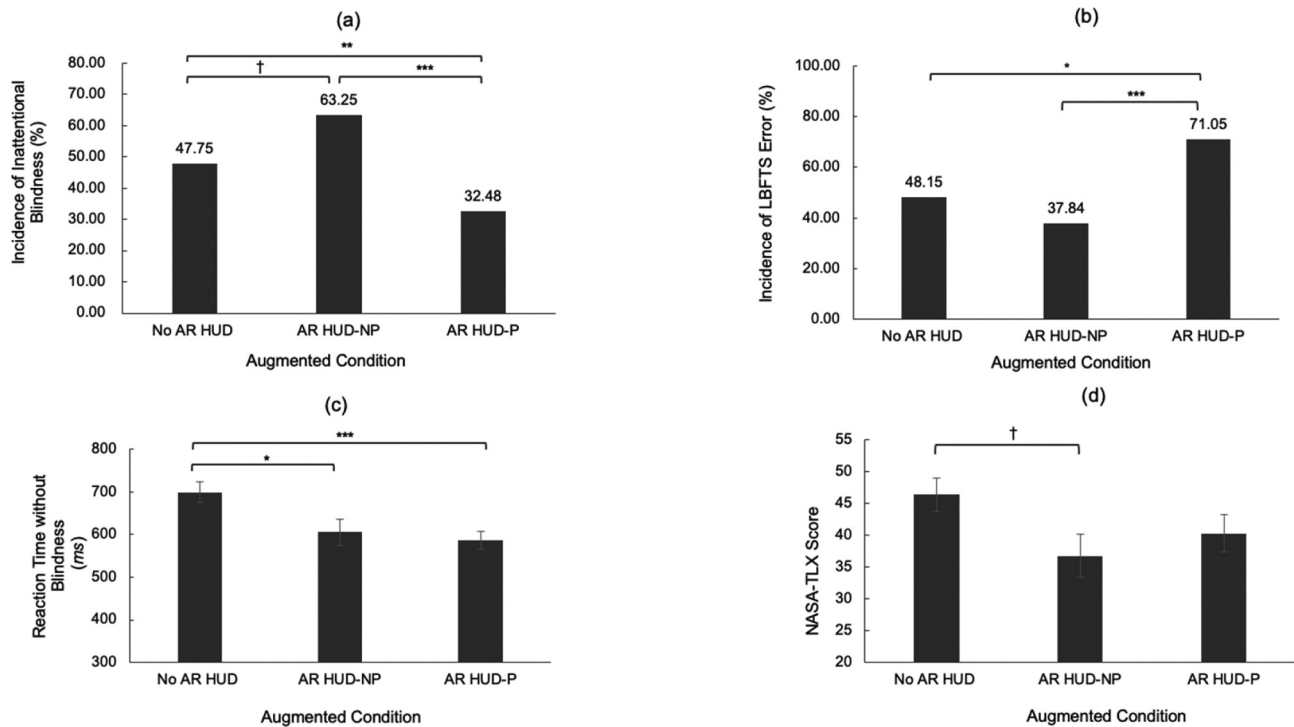


Figure 1. (a) Incidence of inattentive blindness, (b) incidence of two types of blindness, (c) reaction time without blindness (mean and standard error), and (d) NASA-TLX score (mean and standard error) under different augmented condition.

used binary logistic regression as the link function. The result showed that the augmented condition significantly impacted the type of inattentive blindness ($\chi^2(2) = 10.493$, $p < .01$). The GLM parameter estimates results showed that compared with the AR HUD-P condition, the incidences of LBFTS error were significantly lower in No AR HUD and AR HUD-NP conditions ($\chi^2(1) = 4.303$, $p < .05$; $\chi^2(1) = 10.487$, $p < .001$) (Table 3). We conducted the GLM post hoc test to compare the incidence of LBFTS error between AR HUD-NP and No AR HUD conditions (Bonferroni correction). We observed no difference between AR HUD-NP and No AR HUD conditions ($p > .05$) (Figure 1b). Notably, although the proportion of LBFTS errors in the AR HUD-P condition was significantly higher than that of the other two groups. However, the frequency of LBFTS in the three groups was very similar ($n(\text{No AR HUD}) = 27$, $n(\text{AR HUD-NP}) = 28$, $n(\text{AR HUD-P}) = 27$). Thus, the frequencies of ordinary blindness in the No AR

HUD ($n = 28$) and the AR HUD-NP ($n = 46$) conditions were significantly higher than those of the AR HUD-P ($n = 11$) condition.

2.3.4. Reaction time without blindness

We used reaction time without blindness as the dependent variable and the augmented condition as the factor to perform GLM. We used the linear model as the link function. The result showed that the augmented condition significantly impacted reaction time without blindness ($\chi^2(2) = 13.490$, $p < .001$). The GLM parameter estimates results showed that compared with the AR HUD-P condition, the reaction time was significantly longer in the No AR HUD condition ($\chi^2(1) = 12.632$, $p < .001$). No difference was observed between AR HUD-P and AR HUD-NP conditions ($\chi^2(1) = 0.268$, $p > .05$) (Table 3). The GLM post hoc test results (Bonferroni correction) showed that reaction time was

significantly longer in the No AR HUD condition compared with the AR HUD-NP condition ($p < .05$). However, no difference existed between the two AR HUD conditions ($p > .05$) (Figure 1c).

2.3.5. Subjective workload

We used the NASA-TLX score as the dependent variable, the augmented condition as the factor to perform GLM. We used the linear model as the link function. The result showed that the impact of the augmented condition on subjective workload was of marginal significance ($\chi^2(2) = 5.630, p = .06$). The GLM parameter estimates results showed that no significant difference existed among No AR HUD, AR HUD-NP, and AR HUD-P conditions in terms of subjective workload ($ps > .05$) (Table 3). The GLM post hoc test (Bonferroni correction) showed that the subjective workload in the No AR HUD condition was marginally significantly higher than the AR HUD-NP condition ($p = .057$). No difference was observed between the two AR HUD conditions ($p > .05$) (Figure 1d).

3. Experiment 2

3.1. Objectives

The results of experiment 1 showed that when using AR HUD, the participants were more likely to notice augmented stimuli. However, they were more likely to experience inattentional blindness to unpredictable stimuli not augmented. Nevertheless, existing AR HUD designs purposed different strategies over the choice of which stimulus to augment. Moreover, in an intricate driving environment, no AR HUD system can ensure that every safety-critical stimulus is augmented reasonably and promptly. Therefore, inattentional blindness induced by AR HUD failing to augment unpredictable pedestrians urgently needs attention and solution. In experiment 2, we explored how the workload affects the occurrence of inattentional blindness in driving in AR HUD without the augmented unpredictable pedestrian (AR HUD-NP) condition.

3.2. Method

3.2.1. Participants

A total of 20 licensed drivers (mean age = 21.63 years) with normal to corrected-to-normal vision were recruited as participants for this experiment. Their average driving experience

is 2.74 years ($SD = 1.21$) and their average mileage is 283 km. All participants provided written informed consent and were compensated for their participation.

3.2.2. Experimental equipment

The experiment equipment and environment are identical to those in experiment 1.

3.2.3. Materials

In experiment 2, we created five low-workload trials and five high-workload trials on STISIM3. A total of 10 videos were generated through video recording as a licensed experimenter completes all trials in the driving simulators. Trials of different workloads were different in terms of the density of road elements. For example, in a low-workload trial, there were no curves, 1 decelerating lead vehicle, 5 buildings, 20 trees, and 5 pedestrians, while in a high-workload video there were 4 curves, 1 decelerating lead vehicle, 15 opposite vehicles, 20 buildings, 35 trees, and 20 pedestrians (see Figure 2 for demonstration graphs for two workload conditions). Participants would experience six pedestrians-crossing-road events (three in each workload condition). Other settings are the same as those in experiment 1.

3.2.4. Experimental variables and design

This experiment used a within-subject design. The workload condition was the independent variable with two levels: (1) low-workload condition; (2) high-workload condition. All participants viewed low-workload and high-workload videos. The workload can be manipulated by the complexity of the road according to Paxion et al. (2015), such as the number of curves, traffic flow, number of pedestrians, and the complexity of the surrounding environment. We used the NASA-TLX scale scores to verify whether the road complexity is successfully manipulated. The dependent variables were the occurrence of inattentional blindness and reaction time without blindness, which is the same as that in experiment 1. In experiment 2, we did not include LBFTS error as the dependent variable because we did not find any difference in the frequency of LBFTS error among three augmented conditions. Thus, we believed that the incidence of LBFTS error was not a sensitive indicator in our study.

3.2.5. Procedure

The experiment procedure was identical to that in experiment 1. In experiment 2, all subjects completed both high load

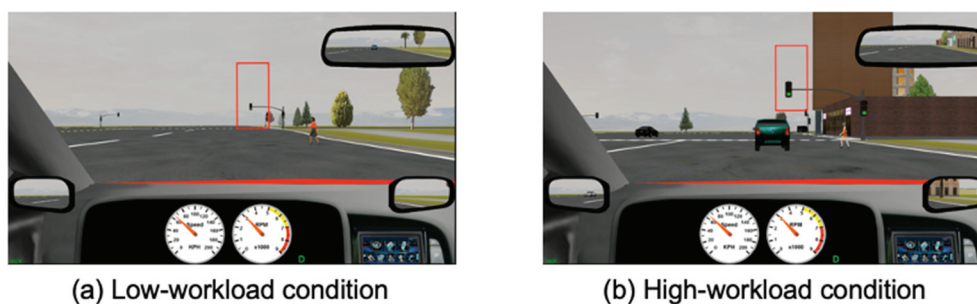


Figure 2. Demonstration graphs for (a) low- and (b) high-workload conditions.

block and low load block., and each block included five trials (i.e. five videos). The presentation order of two blocks was counterbalanced in an ABBA order. That is, half participants followed the “high → low” order, and the other half followed the “low → high” order. We used the Latin square design to counterbalance the sequence of five trials in each block. Participants were instructed to press the button when they noticed the following: (1) the ego-car exceeded the speed limit, (2) the ego-car ran a red light, (3) the distance between the ego-car and the lead car was too close, (4) a participant in front was crossing the road. However, only the data of participants responding to pedestrians were used for analysis.

When participants completed one trial, the next trial automatically started according to the system setting. After completing one block, the participants were asked to complete the NASA-TLX scale to measure their workload of each workload block.

3.3. Results

3.3.1. Data processing

The data processing method was the same as that in experiment 1. The occurrence of inattention blindness is based on drivers' response (response or no response). We coded the result as no blindness if participants responded to unpredictable pedestrians and inattention blindness if they responded. The reaction time without blindness equals the exact time the participants press the brake pedal minus the exact time the unpredictable pedestrian showed up. For the NASA-TLX score, we asked the participants to finish two parts of the questionnaire. We multiplied the weighted score (the score derived from the second part of the questionnaire) and the scale score (the score derived from the first part of the

questionnaire). Then, we divided it by 15 to get our final workload score.

3.3.2. Subjective workload and road complexity

A total of 117 valid trials were obtained. The generalized estimating equation (GEE) was adopted to analyze the occurrence of blindness, the subjective workload, and reaction time without blindness data. Similar to GLM, GEE can handle discrete and non-linear data. GEE is suitable for a within-subject design while GLM is suitable for a between-subject design.

We used the NASA-TLX as the dependent variable and the workload condition as the factor to perform GEE. We used the linear model as the link function. The result showed that workload condition had a significant impact on subjective workload ($\chi^2(1) = 10.556, p < .001$) (Table 4), indicating that we successfully manipulated participants' subjective workload by altering the complexity of roads (Figure 3c).

3.3.3. Occurrence of inattention blindness

We used the occurrence of inattention blindness as the dependent variable and the workload condition as the factor to perform GEE. We used binary logistic regression as the link function. The result showed that the workload condition had a significant impact on subjective workload ($\chi^2(1) = 6.050, p < .05$) (Table 4), and the incidence of inattention blindness was significantly higher in the high-workload condition (Figure 3a).

3.3.4. Reaction time without blindness

We used the reaction time without blindness as the dependent variable and the workload condition as the factor to perform GEE. We used the linear model as the link function. The

Table 4. Effect of workload on inattention blindness and reaction time.

Dependent Variable	Workload Condition	B	SE	Wald χ^2	p	OR	95% CI OR
The occurrence of inattention blindness	High	0.740	0.301	6.050	.014*	2.097	[1.162, 3.782]
	Low						
Reaction time without blindness	High	-141.450	66.126	4.576	.032*	3.71E-62	[1.92E-118, 7.17E-06]
	Low						
NASA-TLX Score	High	-6.267	1.9288	10.556	.001***	0.002	[4.33E-05, 0.083]
	Low						

* $p < .05$; ** $p < .01$; *** $p < .001$.

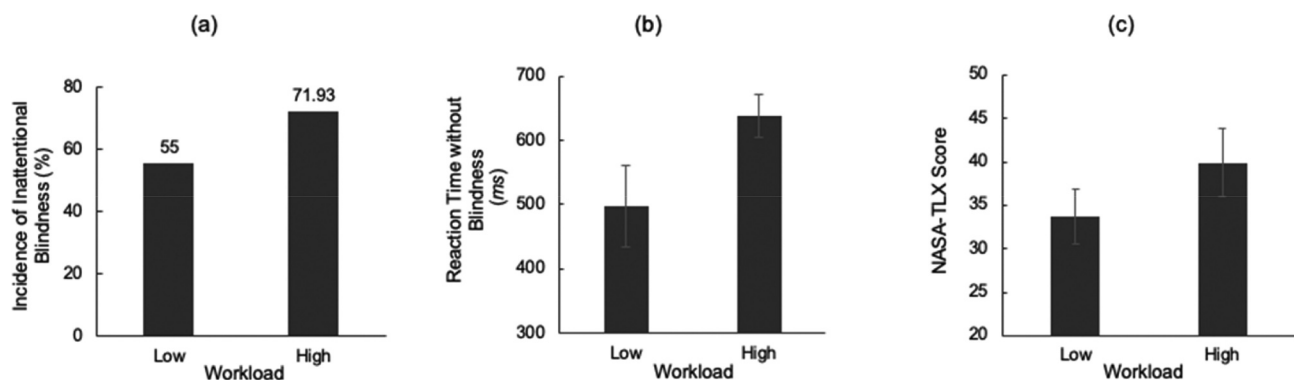


Figure 3. (a) Incidence of inattention blindness, (b) reaction time without blindness (mean and standard error), and (c) NASA-TLX score (mean and standard error) under different workload conditions.

result showed that the workload has a significant impact on the reaction time ($\chi^2(1) = 4.576, p < .05$) (Table 4), and the driver's reaction time was longer in the high-workload condition than in the low-workload condition (Figure 3b).

4. General discussion

The empirical study focused on the inattention blindness in AR HUD-assisted driving. This study compared the occurrence of inattention blindness in different AR HUD-augmented conditions and further explored the impact of workload on inattention blindness. The results showed that utilizing AR HUD could significantly reduce driver's inattention blindness if the unpredictable stimuli were augmented with a bounding box. However, AR HUD could also encourage inattention blindness if the system failed to augment the unpredictable stimuli, and high-workload participants would experience more inattention blindness.

4.1. Impact of AR HUD-augmented condition on inattention blindness

The present study provided empirical evidence that AR HUD in driving could induce inattention blindness toward a subject not augmented, which consists of the conclusion drawn by previous studies (Gabbard et al., 2014; H. Kim & Gabbard, 2019; Oh et al., 2016; Romer et al., 2014). The present study further studied inattention blindness toward unpredictable pedestrians. This study found that drivers using AR HUD were more likely to pay attention to augmented objects, thereby neglecting the safety-critical objects not detected and augmented by the system. That is to say, if the system fails to augment an abruptly appeared pedestrian and instead augment a traffic sign with a bounding box, then the driver may experience inattention blindness toward pedestrians because the drivers' attention is shifted to the augmented traffic sign. This event could lead to serious traffic accidents. The underlying reason may be that the AR bounding box used to augment environmental elements in AR HUD guided the participant's attention to a certain subject, causing them to ignore other stimuli.

We can also explain the result in terms of the theory of attention set. Pammer et al. (2015) first proposed the concept of "attention set" in their study of inattention blindness in driving. Attention set refers to an individual's mental template of objects, situations, or behavior based on its consistent semantic features and feature relationships. Koivisto and Revonsuo (2008) proved that if individuals were faced with multiple stimuli simultaneously, then they would only pay attention to the target stimulus while ignoring others. In other words, individuals built an "attention set" of the features of this specific stimulus. According to the theory of "attention set," the observer establishes an attention set that contains features of the target object to better monitor the environment, enabling the easy detection of stimuli containing the same features. Hence, the more the stimuli overlap with the attention set, the easier it will be noticed by observers. With regard to the driving task, the attention set should have only contained features of objects and processes related to the driving environment. However, if participants use AR HUD, then the "AR bounding box" becomes a new feature of

the participants' attention set. Therefore, the overlap ratio of the augmented unpredictable stimuli and the attention set would increase compared with when not using AR HUD. Meanwhile, the overlap ratio between the unpredictable stimuli not augmented, and the attention set would decrease. This event explains why participants using AR HUD were more likely to detect the augmented stimuli and showed more inattention blindness to the unpredictable stimuli not augmented.

The limitation of the human visual system could also be the underlying reason for AR HUD-induced inattention blindness. According to Treisman's (1982) theory of perceptual grouping, our visual attention is a limited pool of resource which curtails parallel information processing. That is, one cannot focus on two different perceptual groups at once. The theory echoes Wicken's viewpoint that the visual system is limited in monitoring multiple displays for it only has a single focal point (Baldwin et al., 2012; Wickens, 1984, 2002). In a HUD display study, McCann et al. (1993) demonstrated that our visual system parses HUD display as separate perceptual groups. Thus, when drivers focused their attention on HUD display (i.e. the bounding box in the present study), real-world road elements (i.e. The pedestrians) were excluded from processing and vice versa. As a result, drivers need to shift attention constantly between the HUD and the real world, which causes inattention blindness. Although McCann's et al. (1993) conclusion was based on the screen-fixed AR HUD design, we presume that world-fixed AR HUD may contain a similar flaw that induces inattention blindness. Future study is needed to test this hypothesis.

Inattention blindness could also be the result of lacking cognitive resources. Cognitive load theory suggests that our cognitive resources are limited. If the ongoing task demands considerable cognitive resources, then the cognitive resources left are insufficient to process other tasks (Lavie, 1995, 2005). In experiment 1, we found that AR HUD display cannot effectively reduce participants' subjective workload. This notion could explain why drivers under No AR HUD and AR HUD-NP conditions shared a similar proportion of inattention blindness toward unpredictable pedestrians. In experiment 2, we found that high-workload drivers were more likely to experience inattention blindness, and they reacted significantly slower to unpredictable pedestrians compared with low-workload drivers. That is, if the driving task is demanding, then driving would take up much of the driver's cognitive resource, leaving the driver with insufficient cognitive resources to handle unpredictable contingencies.

4.2. Impact of workload on inattention blindness when using AR HUD

Experiment 2 was based on a previous paradigm and explored the impact of workload on inattention blindness in AR HUD-assisted car driving (Murphy & Greene, 2015, 2016). The result confirmed that the high-complexity road environment induced a higher workload, indicating that our manipulation of workload was effective (Paxion et al., 2015). We also concluded that a higher workload would lead to more inattention blindness or prolonged reaction to unpredictable stimuli, which was in accordance with the findings of previous

studies (Hyman et al., 2010; Murphy & Greene, 2015, 2016). Many cognitive studies of inattention blindness also come up with the same conclusion and can help to explain this phenomenon (Cartwright-Finch & Lavie, 2007). For example, Macdonald and Lavie (2008) adopted the classic letter-search paradigm to study inattention blindness. They proposed that load-induced blindness was caused only by the allocation of limited capacity attention, and not by other cognitive processes. Their inference was consistent with cognitive load theory, which believes that our cognitive resources are limited, and when exhausted, we can no longer process other stimuli. That is, we need sufficient cognitive resources to notice and respond to target stimuli in time (Lavie, 2005; Lavie et al., 2014). Therefore, we deemed that drivers failed to detect unpredictable stimuli or to respond to them quickly under high workloads in our experiment. The reason is that the high-complexity road demands considerable cognitive resources from drivers, leaving them with insufficient resources to deal with new stimuli.

Experiment 2 provides additional insight on the impact of workload on inattention blindness in AR HUD-assisted driving. As the incidence of inattention blindness increase and the response time to unpredictable stimuli also increases, indicating that cognitive resource was closely associated with inattention blindness. Thus, future studies should examine their underlying relationship in driving.

4.3. Implications for AR HUD design

World-fixed AR HUD was proved to be more effective than screen-fixed AR HUD as the former is conformal to the real world and will not cluster real-world view. However, the present study revealed that the world-fixed AR HUD could also compromise attention allocation of drivers, that is, the bounding box used to augment important road objects may serve as an attention pointer that attracts too much attention from drivers to augmented objects leaving critical road events undetected. In other words, drivers will be more likely to experience inattention blindness to critical road events under AR HUD assistance. This case counters the original purpose of developing the AR HUD system to reduce road accidents. Moreover, once the drivers realize that the system is overwhelmingly attention-demanding, they may stop using it at all.

Geometric elements such as quadrangle- and triangular-shaped bounding graphics are widely accepted in the AR HUD warning system design (Gabbard et al., 2014). However, few studies have pointed out its limitation. The inattention blindness phenomenon observed in the present study provides a new insight into world-fixed AR HUD design. Cues similar to bounding boxes are too prominent for drivers when they are superimposed on real-world objects. Thus, softer and less distracting cues should be introduced in the AR HUD design to attract an appropriate amount of attention from drivers without negatively affecting their ability to observe the surrounding environment. For example, Kim and Gabbard (2019) have developed “virtual shadow” AR HUD cues that can guide the drivers’ attention without narrowing it. Park and Kim (2013) used a green arrow at the pedestrian’s head position as cues to guide drivers’ attention

properly. In general, we recommend AR HUD cues that only give hints of the existence of augmented (e.g., an arrow) objects instead of extracting them from the real-world environment as the bounding boxes do.

We also recommended using different cues to augment different types of road elements. In our experiments, we used the same bounding boxes to augment pedestrians, traffic lights, and road signs. These road elements often appeared at the same time in the driver’s sight. According to “attentional set” theory, the bounding boxes become the new features of augmented objects in the AR HUD augmented condition, which requires more attention resources for drivers to identify the type of augmented objects and judge their safety significance respectively. If different cues were applied in the AR HUD design, the driver’s mental workload may drop and may develop a safer attention allocation strategy. Previous studies have strived to design diverse AR HUD cues. Park and Kim (2013) proposed an AR HUD interface design with various cues to augment different road elements. Gabbard et al. (2014) summed up different AR HUD cues used in a previous study. S. Kim and Dey (2016) included visual icons in their AR HUD designed to better present road information. We also believe that affective AR HUD design may attract drivers attention naturally and could provide benefit to road safety (Ho & Spence, 2013). Future studies can give some insight into it.

4.4. Limitations and research prospect

The present study has certain limitations. First, the present study chose driving videos as the experimental material in consideration of the paradigm of previous studies, the difficulty of data collection, and the issue of technical limitation (H. Kim & Gabbard, 2019; Lu et al., 2020; Schwarz & Fastenmeier, 2017). Watching video clips and pretending to drive cannot precisely imitate the workload nor the close loop control in real-world driving, leaving the study with low ecological validity. Future studies can repeat the present study in a natural or simulated driving condition to explore the influence of AR HUD on inattention blindness and driving performance. Second, the present study used a portable eye tracker to collect eye movement data, so the participants were carefully instructed to restrict their head movement for data collection. However, keeping their head still in a long driving task is impossible, and this case may eliminate the precision of eye movement data. Moreover, restricting head movement would also hurt the ecological validity of the study. Future studies should introduce more precise apparatus to avoid these problems.

To optimize the existing AR HUD design, additional factors affecting inattention blindness in AR HUD need to be investigated. Previous studies have suggested that the shape and color of unpredictable stimuli affect the occurrence of inattention blindness (Hyman et al., 2010; Simons & Chabris, 1999). With regard to the design of the AR HUD interface, the effect of some properties of AR warning graphic, such as the shape and color and appropriate timing on inattention blindness in driving is still an open question (H. Kim & Gabbard, 2019). Moreover, the present study only focused on whether unpredictable stimuli were augmented in AR HUD.

Other design factors such as the instrument panel, real-time navigation information, and warning information, may also affect inattention blindness in driving. Furthermore, subject variables such as driver's age, experience, and road familiarity may interact with the AR HUD interface and driving environment, causing inconsistent results. Therefore, future studies could focus on these factors to offer new insight into inattention blindness in driving. With the development of AR technology, the multimodal AR technology that integrates multiple sensory channels has been applied to the driving field (Kaczmarek, 1995; Kern et al., 2009; Hwang & Ryu, 2010; Carton & Dunne, 2013; S. Kim & Dey, 2016). A study has proved that the multi-model AR technology can bring better driving performance. However, such technology may also overload drivers with its rich information, causing more inattention blindness. Thus, the application of multi-model AR technology in driving and its impact on inattention blindness still needs further exploration.

5. Conclusion

The present study focused on the inattention blindness in AR-assisted driving and the impact of workload when the AR system fails to augment the unpredictable stimuli. The study found that AR HUD can reduce inattention blindness toward augmented subjects (subjects inside the AR bounding box) while inducing inattention blindness toward subjects outside the AR bounding box. Moreover, a higher workload will intensify inattention blindness toward subjects outside the AR bounding box.

Disclosure of potential conflict of interest

No potential conflict of interest was reported by the author(s).

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About the Authors

Yuwei Wang is a graduate student at the Department of Psychology, Zhejiang Sci-Tech University. Her research interests include traffic safety, user experience design, human–computer interaction.

Yimin Wu is an undergraduate student at the Department of Psychology, Zhejiang Sci-Tech University. His research interests include user experience and human–computer interaction.

Cheng Chen is a graduate student at the Department of Psychology, Zhejiang Sci-Tech University. Her research interests include augmented reality and human–computer interaction.

Bohan Wu is a graduate student at the Department of Psychology, Zhejiang Sci-Tech University. His research interests include user experience design and ergonomics.

Shu Ma is a PhD researcher at the Department of Psychology, Zhejiang Sci-Tech University. Her research interests include human–computer interaction, traffic safety and automated driving.

Duming Wang is an assistant professor at the Department of Psychology, Zhejiang Sci-Tech University. His research interests include human–computer interaction, human factors, and user experience.

Hongting Li is a professor at the Department of Psychology, Zhejiang Sci-Tech University. His research interests include human–computer interaction, ergonomics, user experience, augmented reality and virtual reality.

ZhenYang is an assistant professor at the Department of Psychology, Zhejiang Sci-Tech University. His research interests include human computer interaction, human factors, user experience, cognitive science and neuroergonomics.